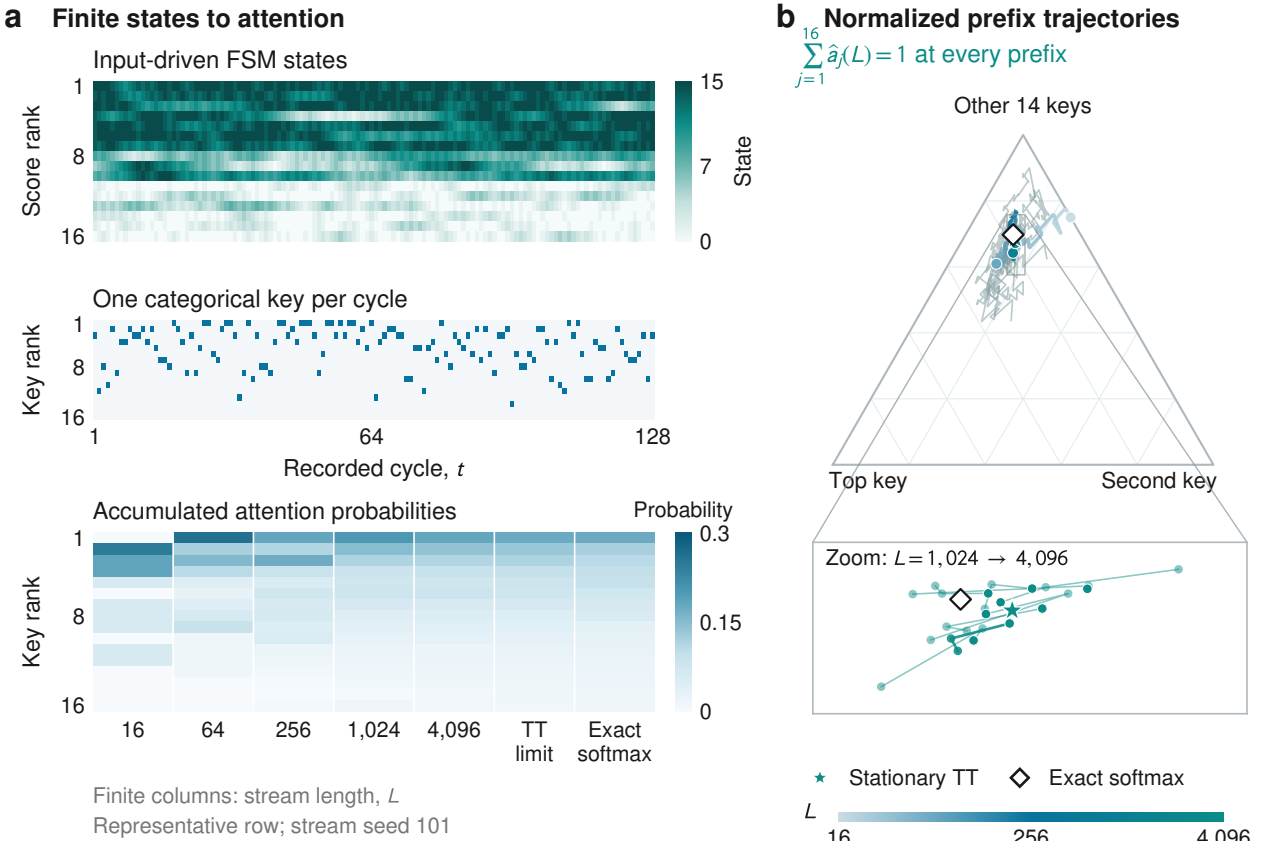

a
Finite states to attention
Input-driven FSM states
Score rank
1
8
16
15
7
0
State
One categorical key per cycle
Key rank
1
8
16
1
64
128
Recorded cycle, t
Accumulated attention probabilities
Probability
Key rank
1
8
16
0.3
0.15
0
16
64
256
1,024
4,096
TT limit
Exact softmax
Finite columns: stream length, L
Representative row; stream seed 101
b
Normalized prefix trajectories
$\sum_{j=1}^{16} \hat{a}_j(L) = 1$ at every prefix
Other 14 keys
Top key
Second key
Zoom: L = 1,024 → 4,096
Stationary TT
Exact softmax
L
16
256
4,096